

Virtual ADM3A Terminal

June 8, 2026

Table of Contents

Running.....	1
Telnet version.....	1
Tty version.....	1
Controls.....	1
Telnet version.....	2
Tty version.....	2
Keyboard.....	2
Properties.....	2
Terminal Option Properties.....	3
Telnet Properties.....	4
Tty Properties.....	4

Running

There are example “bash” scripts showing how to run the telnet and tty (serial port) variants. Serial ports requires the external package jSerialComm which is a submodule in git repository. The general commandline syntax is:

Telnet version

```
java -jar ADM3Atelnet.jar [prop=value...] [host port]
```

Tty version

```
java -cp ${JARS} ADM3Atty [prop=value...] [tty baud]
```

Where “\${JARS}” here represents the paths to the jar files required to run, specifically ADM3Atty.jar and jSerialComm-2.9.3.jar, separated by a colon (Linux/MAC) or semi-colon (Windows).

Controls

Text on the screen may be selected using the left mouse button and dragging the usual way. Text remains selected until it scrolls off the screen or another mouse button click.

The menu item **Screen ► Copy**, or key Alt-C, copies the current selection to the clipboard. This is generally done automatically when the mouse button is released, but may be used to re-copy the current selection if the clipboard contents gets replaced by use outside of the terminal.

The menu item **Screen ► Paste**, or key Alt-V, copies the current contents of the clipboard to the terminal output (sends to the computer). The text is sent at the rate specified by the adm3a_delay property.

The menu item **Screen ► Copy_All**, or key Alt-A, copies the entire current screen text to the clipboard.

Telnet version

The menu item **Screen ► Reconnect** will attempt to reconnect to the computer using the same host and port previously specified, provided the current connection was lost.

The menu item **Screen ► Hangup** will disconnect the terminal from the computer (release the telnet connection).

The menu item **Screen ► New_Connection** will prompt for a new host and port and attempt to connect. This is done after doing a hangup of any current connection.

Tty version

The menu item **Screen ► New_Connection** will prompt for a new tty and baud and then attempt to close and re-open the serial port.

Keyboard

F12 functions as the HERE IS key. The answer-back drum defaults to no characters (empty ROM), but may be set to a string using the "ansbak" property (32 characters maximum). The data is sent using the same delay as for paste operations.

F1 will cancel any current paste or HERE IS transmission being done. This is especially useful if an excessively large amount of text is accidentally pasted.

The Enter key produces CR, use Ctrl-J or Ctrl-Enter for LF.

Delete or Shift-Backspace sends the RUB code (DEL/Rubout) (0x7f)

The cursor keys send the equivalent cursor movement codes.

The Home key sends the HOME cursor code.

The Shift-Delete key sends the CLEAR (screen) code.

Properties

Properties with the "adm3a_" prefix may be specified on the commandline by removing the prefix. Quotes around values may be required, depending on your shell.

The terminal will search for properties in the following order:

1. *prop=value* arguments on the commandline
2. File specified in the environment variable ADM3A_CONFIG
3. Looking for "adm3a.rc" in the current directory
4. Looking for ".adm3arc" in the user's home directory

Only one properties file will be used - they are not combined. Commandline properties override any loaded from a file.

adm3a_ansbak=string

Specifies a string to use for the Answer-back ROM. Typically, this ROM generated no characters

unless programmed, and that is the default. A maximum of 32 characters is used. Only the number of characters programmed are sent. This string may include the standard "escapes" for special characters, such as "\r" for CR and "\n" for LF.

adm3a_delay=ms

The delay inserted after each character is pasted or otherwise automatically sent to the host, in milliseconds. Default is 100.

adm3a_cr_delay=ms

The delay inserted after CR is pasted or otherwise automatically sent to the host, in milliseconds. Default is 1000.

adm3a_font=file

Specifies the font to use for the display. Generally, only the built-in font "fonts/ADM3A.ttf" will work properly, as the character codes need to match what is used internally to select the control character glyphs and reverse-video glyphs.

adm3a_font_size=points

Specifies the font size in points to use for the display. The optimal value for the built-in font is "18", the default.

adm3a_font_color=RGB

Specifies the font color to use for the display. The value is specified in RGB hexadecimal, for example green would be "00ff00". The default is green.

adm3a_beep=wav-file

Specifies the audio clip file to use for the terminal beep. Default is the built-in "adm3a_beep.wav".

adm3a_beep_volume=percent

Specifies the audio volume to use for the terminal beep, in percent. Default is "50".

adm3a_loopback

This is a special mode that sends all keys pressed back to the terminal, bypassing any connected computer. It cannot be turned off without restarting the terminal (without the property being set). It is generally only useful for test purposes. No value is required.

Terminal Option Properties

The following properties accept the values (case-insensitive) "yes", "true", "on", or "1" to enable an option. All other values disable. If the property is not set, the default value is used.

adm3a_auto_nl

Enables the automatic CR-LF at column 80. Default is on.

adm3a_uc_only

Forces all key presses to upper case. Default is off.

adm3a_uc_disp

Forces all characters displayed to upper case. Default is off.

adm3a_clr_scr

Enables the CLEAR SCREEN function. Default is on.

adm3a_ndspace

Enables the non-destructive space option. Not implemented. Default is off.

adm3a_pe

Enables generation of parity in the 8th bit of key codes sent. Default is off.

adm3a_ep

Selects the state of the 8th bit of keyboard codes. Default is on when parity is enabled (EVEN parity), off when parity disabled (SPACE parity bit).

adm3a_col72_beep

Enable/disable the beep at column 72. Default is enabled.

Telnet Properties

adm3a_host=*host*

The hostname or IP address to connect to. This is the same as the first non-property argument on the command line.

adm3a_port=*port*

The port number to connect to. This is the same as the second non-property argument on the command line.

Tty Properties

adm3a_tty=*device*

The hostname or IP address to connect to. This is the same as the first non-property argument on the command line.

adm3a_baud=*speed*

The port number to connect to. This is the same as the second non-property argument on the command line.